When Threshold Sampling Reverses VUS Rankings: Exact and Certified Evaluation of Anomaly Scores

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Abstract

Volume Under the Surface (VUS) compares anomaly scores across temporal tolerance widths and score thresholds. Its mathematical definition can be threshold-independent while a finite integration grid remains inaccurate. We analyze a specific failure of the original dynamic VUS-PR: soft-label mass admitted after exhaustive recall saturates can inflate the precision assigned to an earlier recall jump. We give a finite overestimation bound exceeding 0.98 and same-label, distinct-score constructions at 250 thresholds with exact-score selection regret of 0.507, 0.701 and 0.846 at maximum widths 30, 100 and 300. We derive bidirectional error enclosures and implement them with integer square roots and directed integer arithmetic, certifying the ideal real-arithmetic target without relying on a floating-point tolerance. A prospectively selected panel re-evaluates 1,536 published detector score vectors on 128 time series from sixteen source families. At width 30, the default grid selects a model with exhaustive regret at least 0.01 on eight series across four families; the largest regret is 0.399. Integer enclosures prove all eight regret claims and the unique exhaustive winner on all 128 series. Partial grids certify 63 winners at 250 thresholds and 94 at 1,000. All stored score vectors have at most 101 distinct values: requesting more ranks than distinct scores does not ensure exhaustive integration. We report source-overlap sensitivity, signed errors, matched exhaustive controls, and certification costs.

1 Introduction

An anomaly detector produces scores; an evaluation procedure turns those scores into evidence for comparing or selecting models. Approximation in the evaluator can therefore affect the scientific conclusion even when the labels, models, and score vectors do not change. Volume Under the Surface (VUS) extends range-aware ROC and PR areas by averaging across temporal tolerance widths (Paparrizos et al., 2022; Boniol et al., 2025). This work studies fidelity to one explicitly pinned VUS-PR definition, rather than whether that metric is the best proxy for a deployment's utility.

Finite-threshold approximation is well known. Classical work shows that PR interpolation differs from ROC interpolation (Davis & Goadrich, 2006), and weighted PR requires additional care (Keilwagen et al., 2014). Keras documents finite-threshold errors and bounding estimates (Keras contributors, 2026); TimeEval also warns about coarse VUS threshold grids (TimeEval contributors, 2026). PAULA derives sampled-rank precision bounds under explicit monotonicity assumptions in an annotation-budget setting (Sabharwal & Sedghi, 2017). Generic claims that approximation loses information, or that evaluation metrics can change a ranking, would add little to this literature.

Our specific observation concerns a dynamic soft-label target in which recall is clipped before all soft positive mass has entered. Precision can continue increasing after recall reaches one. A sampled threshold can assign this later precision to a recall jump that happened at a much lower precision. The corresponding exhaustive integral assigns zero area to the post-saturation mass. This produces a qualitatively different mechanism from the familiar loss of early positive ranks under coarse sampling. Figure 1 illustrates the mechanism and large same-label ranking reversals at ordinary maximum widths.

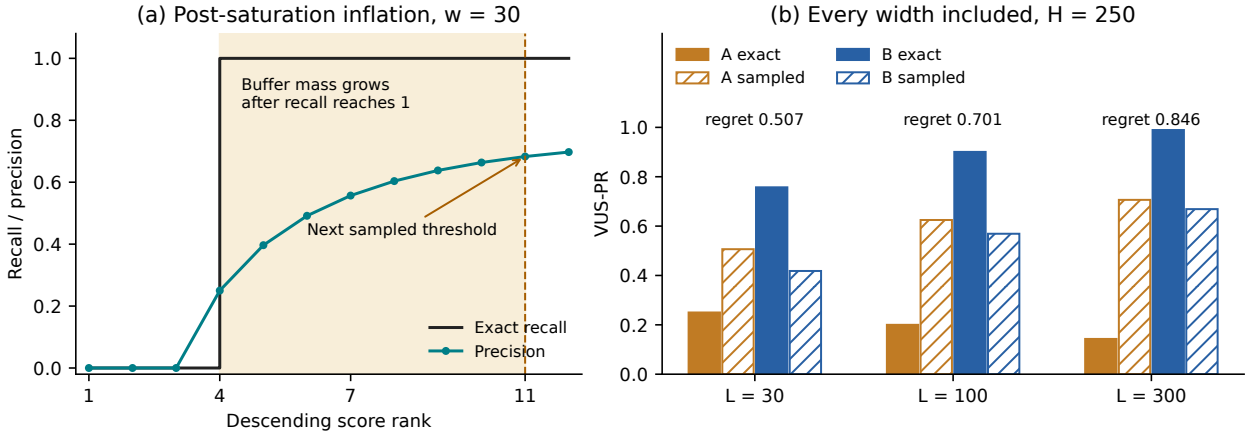


Figure 1: Constructed failures, not prevalence estimates. Left: after the sole original positive enters at rank four, recall is one while buffer mass keeps increasing precision. The next retained threshold is rank eleven. Right: all-width averages at the standard 250-threshold budget prefer score vector A although exhaustive evaluation prefers B.

We provide three forms of evidence. First, explicit constructions and proofs establish what the approximation can do, including large finite overestimation and selection regret. Second, a mass-envelope analysis yields intervals that can certify the exhaustive ordering of models, with an integer implementation that preserves the mathematical guarantee. Third, a fixed, previously unmeasured bank of twelve released score configurations per series separates worst-case possibility from observed effects and computational costs. The score bank is used for offline evaluation with labels, not label-free selection at deployment.

A publicly released support-event evaluator supplies the numerical exhaustive baseline (Ko, 2026). We do not relabel that earlier implementation, its already measured speedups, ordinary radix sorting, or generic PR error bounds as new discoveries. The new investigation isolates post-saturation inflation, a sound integer remedy, and its consequences on a broader published detector panel. Likewise, certification of a ranking does not certify the detector's real-world usefulness.

2 Sampling can inflate the dynamic PR integral

Fix the original dynamic VUS-PR contract for n binary labels with both classes present, $0 < P_0 < n$, finite scores, and a nonnegative integer maximum width L. The implementation evaluates the scores after its documented binary64 conversion. At width w, original positives have unit weight and each interval adds a square-root ramp $\sqrt{1 - d/w}$ at distances $1 \le d \le \lfloor w/2 \rfloor$; overlapping contributions are summed and clipped at one. At a complete score threshold, let C be the predicted count, A the admitted original positive count, B the admitted buffer mass, P_0 the total original positive count, and h/K_w the fraction of expanded components hit. Overlapping expanded intervals merge, whereas adjacent nonoverlapping intervals remain distinct. The coordinates are

$$T = A + B, \qquad P = P_0 + B/2, \qquad R = \min(T/P, 1)h/K_w, \qquad \pi = T/C.$$

The exhaustive PR area is $\sum_j (R_j - R_{j-1})\pi_j$, beginning at $R_0 = 0$ and admitting each distinct global score tie completely. VUS-PR averages this area over every integer width $0, \ldots, L$.

The sampled implementation retains descending zero-based ranks `linspace(0,n-1,H).astype(int)`, with $H = 250$ by default in both the official VUS optimized API (VUS contributors, 2026) and the pinned fast-VUS source (Klein, 2026). It uses the same PR integral on those retained states. Sampling error is different from a change of metric semantics or an implementation defect: the upstream routine correctly computes its configured sampled target.

The important feature is that precision can increase *after* recall has already reached one. Additional buffer mass then contributes zero exhaustive area, but a coarse threshold can incorrectly assign that later precision to a recall jump which occurred earlier. Binary average precision cannot add positive mass after every original positive has already been admitted. The following construction isolates the additional soft-label mechanism.

Proposition 1 (A parameterized sampling family)**.** *Let $H \geq 2$, $1 \leq m < q$, $n = 1 + (H-1)q$, and $\lfloor L/2 \rfloor < n - m$. Put a single original positive at temporal position zero. Assign distinct descending scores first to the last m positions, then to position zero, and then to positions $1, \ldots, n-m-1$ in temporal order. The exhaustive VUS-PR equals $E = 1/(m+1)$, whereas its H-threshold sampled value is*

$$S = \frac{1 + (L+1)^{-1} \sum_{w=0}^{L} B_w}{q+1}, \qquad B_w = \sum_{d=1}^{\min(q-m, \lfloor w/2 \rfloor)} \sqrt{1 - d/w}, \tag{1}$$

where the empty sum at $w = 0$ is zero. The sign of $S - E$ depends on the parameters; the overestimation regimes below are separate conclusions.

Proof. The first m positions are outside every evaluated support. At actual rank $m+1$, admitting the sole original positive makes both coverage and clipped recall one. The entire exhaustive recall increase therefore has precision $1/(m+1)$, independently of width. Later recall increments vanish. The retained one-based ranks are exactly $1, 1+q, \ldots, n$. Recall at the first is zero; recall at the second is one, and the second includes the original plus the nearest $q - m$ positions. Its precision is $(1 + B_w)/(q+1)$, proving the formula after averaging widths. □

Corollary 1 (Large finite overestimation)**.** *For $H = 250$ there is a finite member of Proposition 1 with $S - E > 0.98$.*

Proof. For $L \geq 2q$, retain only widths $2q, \ldots, L$ and use $\sqrt{1-x} \geq 1 - x$ to obtain

$$S \geq \frac{(L - 2q + 1)(q - m + 1) - \frac{(q-m)(q-m+1)}{2} \sum_{w=2q}^{L} w^{-1}}{(L+1)(q+1)}.$$

Take $q = 2{,}000{,}000$, $m = 1000$, $n = 498{,}000{,}001$ and $L = 498{,}000{,}000$. The harmonic tail is less than $\log(L/(2q-1)) < 5$; the latter inequality follows, for example, from the degree-seven positive Taylor lower bound for $\exp(5)$. Thus $S > 977516488501501/996000500000001$. Exact rational subtraction of $E = 1/1001$ exceeds $98/100$. □

This finite statement was verified symbolically, without allocating the large input or executing an evaluator on it. Its maximum width is $n - 1$; it is not a claim about a typical small tolerance width. More generally, with fixed $H > 3$, $L = (H-1)q$, $m \to \infty$ and $m/q \to 0$,

$$\liminf S \geq 1 - \frac{2}{H-1} - \frac{\log((H-1)/2)}{2(H-1)}, \qquad E \to 0.$$

This asymptotic family is a real-arithmetic statement, rather than an unbounded supply of distinct binary64 scores.

Large selection regret at ordinary fixed widths. For a second score vector on the same labels, place a nearby buffer position d first and the original second. Fill the remaining $q - 1$ positions through rank $q + 1$ with far outside-support negatives; rank everything else later. Sufficient feasibility conditions are $n - q + 1 > \lfloor L/2 \rfloor$ and $1 \leq d < n - q + 1$. At $w < 2d$ the exhaustive and sampled areas are respectively $1/2$ and $1/(q+1)$. Otherwise, writing $a = \sqrt{1 - d/w}$ and $r = a/(1 + a/2)$ gives

$$E_B(w) = ra + (1-r)(1+a)/2, \quad S_B(w) = ra + (1-r)(1+a)/(q+1).$$

Table 1 reports complete width averages. All scores are distinct finite integers. Closed forms, independent dense evaluation, compiled exhaustive evaluation, and unchanged pinned upstream calls at $H = 250$ and $H = n$ agree numerically. These are exploratory constructed witnesses of possibility, not estimates of prevalence.

L	n	(q, m, d)	E_A	S_A	E_B	S_B	regret
30	2491	$(10, 3, 5)$	.250000	.506412	.757142	.418181	.507142
100	4981	$(20, 4, 5)$	.200000	.624743	.900580	.569008	.700580
300	9961	$(40, 6, 1)$	.142857	.706289	.989249	.668767	.846392

Table 1: Same-label reversals with $H = 250$. Sampling selects A; exhaustive evaluation selects B. Regret is $E_B - E_A$, measured in exhaustive VUS-PR units and not in an operational detection utility.

3 Sound bounds from retained threshold states

An approximation can be accompanied by a checkable interval instead of being treated as an exact scalar. The general idea of bounding numerical quadrature is established; our result supplies a bound for this dynamic, clipped-recall, soft-label target without assuming monotone precision.

Theorem 1 (Mass-envelope PR enclosure)**.** *At one width, consider two consecutive retained complete-threshold states (c_a, t_a, r_a) and (c_b, t_b, r_b), including an initial zero state and discarding repeated states. Put $\delta = t_b - t_a$ and*

$$\ell = \frac{t_a}{c_b - \delta}, \qquad u = \frac{t_b}{c_a + \delta}.$$

If $c_b - \delta = 0$, set $\ell = 1$; a zero-mass block contributes zero. The exhaustive PR area inside this block is between

$$\int_{r_a}^{r_b} \max(\ell, P_0 r / c_b)\, dr \quad \text{and} \quad (r_b - r_a)u. \tag{2}$$

Summing these bounds over blocks and averaging widths encloses exhaustive VUS-PR. In particular, disjoint model intervals certify their exact order.

Proof. Every temporal weight lies in $[0, 1]$. At an intermediate predicted count c, its true mass t satisfies

$$\max(t_a, t_b - c_b + c) \le t \le \min(t_b, t_a + c - c_a).$$

Dividing by c and optimizing the two continuous piecewise envelopes gives $\ell \le t/c \le u$. Optimizing over continuous counts is safe for integer counts and complete ties because it enlarges the feasible set. If the lower denominator vanishes, all admitted weights are one.

Recall is nondecreasing along descending scores: $(A+B)/(P_0+B/2)$ increases in A and B, with derivative in B proportional to $P_0 - A/2 > 0$. Clipping and multiplication by nonnegative, nondecreasing coverage preserve this property. Thus all recall increments are nonnegative. Also $T \ge P_0 R$, and $C \le c_b$, so contributing precision is at least $P_0 R / c_b$. The exact right-step sum dominates the integral of the nondecreasing lower envelope; its upper bound follows by summing recall increments times u. □

The bound is elementary and can be loose. Refining a retained grid monotonically tightens the real-arithmetic enclosure: $c - t$ is nondecreasing, so each subblock has a lower mass envelope at least as large, an upper envelope no larger, and a no-smaller linear lower envelope. If consecutive global distinct-score ordinals establish that no valid threshold was omitted, use the observed exact right-step term instead. A complete tie can admit many positions without containing any valid intermediate threshold.

An information limit. Endpoint-only bounds cannot always be narrow. Fix $H \ge 3$ and L; choose one original interval of m points and a sampling gap $q \gg m + L$. Two score orders can put those originals immediately after the first off-support negative, or at the end of the same gap, while giving identical (C, A, B, h) at every retained threshold and every width. Their exhaustive areas are obtained by placing the entire maximum-width halo in a common tail after that gap and filling the remaining gap with off-support points. Recall saturates before the common halo enters. The exhaustive areas are respectively

$$\frac{1}{m}\sum_{j=1}^{m} \frac{j}{j+1}, \qquad \frac{1}{m}\sum_{j=1}^{m} \frac{j}{q+1-m+j}.$$

The first tends to one and the second to zero as $m \to \infty$ and $q/m \to \infty$. A bound valid for every score ordering consistent with that retained transcript must include both. This is an ordinary ranking information-loss argument; it does not lower-bound methods that inspect omitted support scores or adapt their thresholds.

4 Integer-directed realization

Theorem 1 concerns real arithmetic. Adding a floating-point tolerance to a numerical check would not implement that guarantee. We instead construct dyadic enclosures using integer square roots and directed integer division.

Let $D = 2^{32}$ be the weight scale. For a single ramp at distance d, set $z = (w - d)D^2$ and $q = \mathrm{isqrt}(\lfloor z/w \rfloor)$. Then $q/D \le \sqrt{1 - d/w} \le (q+1)/D$, with exact upper numerator q when $q^2 w = z$. Original positives and points covered by two ramps have exact numerator D; inactive points have zero. Since every active ramp is at least $1/\sqrt{2}$, two ramps indeed clip to one. For $n < 2^{31}$, all compiled signed-64-bit cumulative buffer sums are below 2^{63}. Nonlinear calculations and final sums use arbitrary-precision Python integers; public NumPy integer parameters are normalized before them.

For an endpoint with exact original count A, buffer brackets $[B_-, B_+]/D$, and exact hit count h, use

$$f(B) = \frac{h \min\{2(AD + B),\ 2P_0 D + B\}}{K_w(2P_0 D + B)}.$$

Monotonicity gives $f(B_-) \le R \le f(B_+)$. With output scale $S = 2^{56}$, floor the lower product $Sf(B_-)$ and ceil the upper product $Sf(B_+)$. Both endpoint precision envelopes can be rewritten as functions nondecreasing in each total-mass argument:

$$\ell(t_a, t_b) = \frac{t_a}{c_b - t_b + t_a}, \qquad u(t_a, t_b) = \frac{t_b}{c_a - t_a + t_b}.$$

Substitute both lower mass endpoints into ℓ and both upper endpoints into u. Combine these rational precision bounds with directed recall differences. When $R_a^+ < R_b^-$, the linear lower refinement is integrated analytically over $[R_a^+, R_b^-]$, a subset of the actual recall interval; otherwise that refinement contributes zero. Floor each lower block numerator, ceil each upper numerator, then apply the same directions after width averaging. No irrational or floating-point operation enters this chain of bounds.

The returned certificate retains integer endpoints and scale. Strict ordering compares integer cross-products, while float displays are rounded outward and are not used for the decision. The certified target is ideal real-arithmetic VUS-PR, not a promise to contain the exact bit pattern of every floating-point summation implementation. An application requiring nested refinement intervals must intersect independently rounded bounds. This implementation is not an adaptive refinement policy.

An independent standard-library calculation checked 14,400 enclosures against exact rational evaluations at deliberately coarse weight/output precisions of three/twelve bits, and 12,500 root brackets by squared-integer inequalities. The production implementation is additionally checked against an independent 90-digit direct-ramp oracle, complete ties, boundary and overlapping intervals, saturated recall, and signed/unsigned NumPy integer parameters. These checks exercise the implementation; the universal guarantee follows from the inequalities and accumulator bounds above.

5 Matched exhaustive and sampled computation

Both routes use the same numerical score conversion, complete global ties, radix preparation primitives, interval geometry and compiled PR kernel. The exhaustive route retains every support event; the sampled route assigns each support event to its first retained global threshold. This matched control avoids attributing a Python-versus-compiled implementation difference to exactness. Both calls include validation and preparation.

For completeness, let S_L be maximum-width expanded support and $Q = |S_L|$. A score group containing no point in S_L changes neither weighted positive mass nor component coverage. Its recall increment and PR contribution are zero. At each retained support score v, its precision denominator must nevertheless be the *global* count $|\{i : s_i \geq v\}|$, including off-support positions tied with it. All nonzero PR contributions are therefore recovered from support events and their complete global counts. Two nearest distinct original intervals suffice to determine every weight: no active ramp gives zero, exactly one gives its square root, and two give one after clipping. Ordered intervals permit a linear temporal scan.

Eight stable byte-radix passes order finite binary64 scores, with signed zeros sharing a numerical tie. Preparing counts and geometry takes $O(n+Q)$ fixed-word operations. Each width costs $O(Q)$, giving $O(n+(L+1)Q)$ exhaustive evaluation with $O(n + Q + L)$ auxiliary words. This is a fixed-word, unit-cost metric-arithmetic bound, not a comparison-model sorting result for arbitrary real numbers or a bound on exact symbolic square-root arithmetic. The floating-point evaluator is numerically exhaustive; only the separate integer certificate encloses the ideal real target rigorously.

The sampled implementation has the analogous $O(n + (L + 1)(Q + H))$ bound before deduplication of retained threshold states. Both scan maximum support. Consequently fewer requested thresholds need not make this matched sampled control faster than the exact sparse route. Certificate costs additionally include bounded-precision integer root brackets and arbitrary-precision endpoint arithmetic; its usefulness and overhead require measurement rather than inference from H alone.

6 Prospective published-score evaluation

6.1 Inputs, fixed comparisons, and independent controls

The MSAD archive supplies twelve stored detector configurations: IFOREST, LOF, MP, NORMA, IFOREST1, HBOS, OCSVM, PCA, AE, CNN, LSTM and POLY (Sylligardos et al., 2023). Before computing metric outcomes, we fixed a family-stratified SHA-256 order over 1,863 metadata-eligible files, excluding training filenames and raw members larger than 12 MB. A fixed round-robin queue and validity/identity rules selected 128 files from sixteen families. Of 148 considered files, nineteen identity duplicates and one single-class file were excluded; no acquisition failure changed the selection. The final panel has 15,793,514 temporal observations and 1,536 score vectors. A separate parser rechecked all 1,664 source members and 1,792 arrays, including 189,522,168 score scalars.

All arrays are preserved without padding, repair, new quantization or outcome-based exclusion. Each stored score vector has at most 101 distinct values; every textual score has at most two decimal places. Thus this is a study of released, quantized scores, not recovered full-precision model outputs. The budget $H = 250$ already exceeds the number of distinct scores, but uniform *rank* sampling can revisit large ties and miss rare score classes. Enumerating every distinct score is a simple, important control.

The primary setting is $L = 30, H = 250$; widths 100 and 300 and budgets 32, 64, 128, 500 and 1,000 are sensitivity analyses. We retain full tied winner sets at tolerance 10^{-10}, and report best- and worst-case exhaustive regret among sampled tied winners. The prospective practical decision uses

$$\widehat{\mathcal{W}} = \{m : s_m \geq \max_j s_j - 10^{-10}\}, \qquad \Delta_{\text{best}} = \max_j e_j - \max_{m \in \widehat{\mathcal{W}}} e_m,$$

where e_m and s_m are the exhaustive and sampled values. Worst-case regret replaces the second maximum by a minimum. The prospective practical decision rule requires best-case regret at least 0.01 in at least 5% of series and at least three source families. This rule is a declared finite-panel study threshold, not a universal utility cutoff or publication criterion. Per-family and equal-family results accompany pooled results; model pairs and widths are not treated as independent observations.

At the primary setting, all 1,536 matched sampled results agree with the unchanged pinned upstream implementation to 1.06×10^{-15}. The fixed first two retained series per family give 30 validation series and 360 score vectors. All agree with both an independent dense exhaustive implementation and unchanged upstream evaluation at $H = n$, with maximum absolute differences 4.45×10^{-15} and 3.34×10^{-16}. All 720

Family	N	errors	max regret	Family	N	errors	max regret
Daphnet	7	0	.0000	MITDB	8	0	.0000
Dodgers	1	0	.0000	NAB	9	0	.0000
ECG	11	0	.0000	OPPORTUNITY	10	0	.0000
GHL	10	1	.0272	Occupancy	2	0	.0000
Genesis	1	0	.0000	SMD	9	0	.0000
IOPS	11	0	.0000	SVDB	10	0	.0000
KDD21	10	2	.3986	SensorScope	10	0	.0000
MGAB	10	1	.2517	YAHOO	9	4	.2362

Table 2: Primary setting on all 128 series. "Errors" counts sampled winner sets whose *best* exhaustive value is at least 0.01 below the exhaustive maximum. Maximum regret includes all cases in each family.

independent exhaustive calls complete without errors or timeouts. These are numerical controls; integer evidence below strengthens the ordering and regret claims to the ideal real-arithmetic target.

6.2 Sampling changes model-selection decisions

At $L = 30, H = 250$, the sampled winner set changes on 10/128 series, including nine changes between unique winners. Eight series (6.25%) in four families have best-case exhaustive regret at least 0.01; the maximum regret is 0.398575. Thus the prespecified finite-panel effect threshold is met. The equal-family practical-error fraction is 5.28%. A 2,000-replicate source-family bootstrap gives a pooled 95% percentile interval of [1.09%, 13.20%]. The observed panel passes the declared threshold; this interval does not establish population prevalence above 5%. Table 2 retains every family, including the twelve with no practical winner error.

Errors have both signs. Across the 1,536 scalar evaluations, mean absolute error is 0.009286, maximum underestimation is 0.560123 and maximum overestimation is 0.161692. Fifteen values overestimate by at least 0.01 and 150 underestimate by at least 0.01. There are 106 strict pairwise reversals, 22 with both exact and sampled margins at least 0.01; these pairs are descriptive, not independent samples. The new inflation mechanism is therefore not a universal explanation for the observed selection errors.

Source-overlap sensitivity. The selection excluded exact label, observation and standardized-observation identities against 362 prior inputs. A subsequent bounded audit additionally compared joint observations and labels after dropping either endpoint. It found 20/128 matches to prior inputs under these transformations. We keep the frozen 128-case primary panel and report the descriptive 108-case nonmatching subgroup: four cases (3.70%) in two families have regret at least 0.01, with maximum 0.236208. This subgroup does not meet the original 5%/three-family threshold. The 20 matched cases contain four practical errors and the overall maximum. The bounded audit neither proves the remaining physical recordings independent nor permits describing the full panel as previously unseen recordings. Its newly evaluated content is the released detector-score bank under this pinned metric contract.

6.3 A measured instance of post-saturation inflation

To connect the construction to actual scores, an exploratory diagnostic uses all twelve models on all eight primary harmful-selection series. For each width it inserts the first exhaustive threshold at which recall reaches one into the original sampled grid, adding at most one threshold. Integer mass brackets independently certify each first-saturation event and its predecessor on all 96 score vectors at all 31 widths. The difference between the original and inserted-grid areas measures the net effect of postponing that saturation boundary; the remaining inserted-grid error measures other omitted-threshold effects. This is an outcome-selected mechanism analysis, not a second prevalence estimate or a claimed faster algorithm.

The largest-regret series, KDD21/005 (the distorted CIMIS air-temperature recording), has 8,183 points and one original positive. Its sampled winner is LOF, while exhaustive evaluation selects CNN (Table 3). For LOF, inserting saturation reduces the area by 0.129093, or 79.84% of its total 0.161692 overestimation. The

Model	Exhaustive	Sampled, $H = 250$	Saturation inserted
CNN	.500000	.055416	.500000
LOF	.101425	.263117	.134024

Table 3: Two unchanged published score vectors on KDD21/005. The diagnostic inserts at most one threshold per width and isolates a particular error component; it does not remove all integration error.

Retained grid	median states	certified winners	median width	max width
250 ranks	50	63/128	6.237×10^{-3}	$\approx .99$
1,000 ranks	68	94/128	3.900×10^{-4}	$\approx .99$
All score classes	96	128/128	2.766×10^{-11}	1.714×10^{-9}

Table 4: Integer enclosures at $L = 30$. State counts and interval widths are per score vector; winner guarantees compare all twelve models jointly. A small median interval width does not replace the all-competitor test.

net change decomposes into a positive 0.181891 buffer-mass term, zero original-mass term and a negative 0.052797 count-dilution term. For CNN, count dilution dominates: the corresponding terms are $+0.026004$ and -0.470588, explaining its 0.444584 underestimation. Inserting saturation recovers the exhaustive winner on three of the eight series; five selections remain wrong. Thus the mechanism occurs in real published scores, but correcting it alone is insufficient in general.

6.4 Integer certificates and a full-grid control

All 3,072 partial-grid certificate calls complete: twelve configurations on all 128 series at $L = 30$, for $H = 250$ and $H = 1,000$. A model is certified as the unique exhaustive winner only if its integer lower endpoint exceeds every competitor's integer upper endpoint. At $H = 250$, 63/128 series (49.22%) satisfy this test; the equal-family proportion is 50.52%. At $H = 1,000$, 94/128 (73.44%) satisfy it, with equal-family proportion 72.87%. All certified winners agree with the corresponding unique sampled winner. No sampled winner set is certified strictly suboptimal *by these partial-grid intervals*. The intervals confirm a subset of choices but leave the erroneous choices unresolved.

The strong baseline retains every distinct score class by requesting $H = n$ and deduplicating complete threshold states. Its 1,536 calls also complete, and certify the unique exhaustive winner on *all 128* series. Compared with the original $H = 250$ selector, they prove ten strictly suboptimal sampled winner sets and eight with regret at least 0.01. The largest guaranteed regret lower bound is 0.3985750301665. At $H = 1,000$, one sampled set is provably suboptimal, with guaranteed regret 0.0018926253; none reaches 0.01. These are exact integer-endpoint decisions, not comparisons to a floating-point reference with a tolerance.

All prior certified winners are preserved, and no full-grid interval is disjoint from its corresponding partial-grid interval. Independently rounded intervals need not nest; real-arithmetic refinement monotonicity is not asserted for the raw rounded outputs. Full-grid certification is an explicit same-guarantee baseline, not an implicit claim that partial certificates are more efficient. Costs are evaluated separately below.

A real target versus a floating-point reference. One $H = 1,000$ interval has the floating-point exhaustive answer 7.63×10^{-16} below its lower endpoint. Direct analysis resolves the case exactly: on the YAHOO/CNN input, three original intervals never merge over widths zero through thirty, eight leading originals cover all components, and the ninth original joins one negative in a complete score tie. Buffer points enter only after recall saturates. At every width the ideal value is $8/9 + (1/9)(9/10) = 89/90$, which the certificate contains. The discrepancy is floating-point accumulation in the reference, not an interval violation. No tolerance or widened interval was used to remove it.

Maximum width	$H = 32$	64	128	250	500	1,000
30 (primary)	21	16	11	8	1	0
100	20	15	12	7	2	0
300	20	15	11	6	2	1

Table 5: Series with best-case exhaustive regret at least 0.01; all denominators are 128. These sensitivity cells are not independent studies.

Comparator / exhaustive radix	geometric ratio	family 95% interval	outputs
Matched compiled, $H = 250$	1.044	[1.029, 1.062]	PR
Saturation exact	1.011	[0.999, 1.026]	PR
Unchanged Klein, $H = 250$	5.077	[4.186, 6.403]	PR
Independent dense exact	13.481	[12.505, 14.583]	ROC and PR

Table 6: 360 paired series/model cases at $L = 30$, five repetitions per method. Ratios above one favor exhaustive radix. The matched compiled row controls for preparation and implementation family.

6.5 Sensitivity and matched computation costs

All 36,864 declared numerical calls finish successfully, including all 4,608 upstream $H = 250$ parity comparisons across the three widths. Table 5 reports every declared budget/width combination. Raising the budget to 1,000 removes practical selection regret at widths 30 and 100 on this panel, while width 300 retains one case with regret 0.095202. At $H = 250$, widths 100 and 300 affect five and three families respectively. Width 300's six cases fall below the prespecified 5% fraction; neither width replaces the primary result.

Timings use the fixed 30-series validation subset, all twelve models, five independently randomized paired repetitions, one compute thread, a 30-second per-call process watchdog and non-study warmup. Other research compute jobs stop for this phase. Every timed scalar call includes public-API validation, preparation and integration. Input decoding, imports/JIT warmup, process setup, winner aggregation and record writing are outside its timer. The broader collection wall time is recorded separately and is not substituted for scalar-call latency. All 9,000 planned timing calls complete with output parity and no timeout.

Table 6 reports per-case median times aggregated as geometric comparator/exhaustive-radix ratios, with source-family bootstrap intervals. Exact computation is modestly faster than matched compiled sampling: the ratio is 1.044, with interval [1.029, 1.062]. The larger ratio against unchanged Klein includes implementation differences and cannot be interpreted as an intrinsic exact-versus-approximate gain. The independent dense control additionally computes ROC internally; its output scope is disclosed rather than used as a matched PR-only speed claim. The saturation-exact interval includes one. These are warm, one-host measurements on the released quantized score panel.

Cost of the stronger guarantee. A separate frozen addendum measures the $H = 250$ integer certificate, the full distinct-score integer certificate, and floating-point exhaustive radix together on the same 360 cases, with five randomized repetitions. The three-method comparison was specified before any timing outcomes; full-grid accuracy outcomes were already known when its final protocol was frozen. The original two-method protocol remains unexecuted and preserved, avoiding duplicate timing selection. All 5,400 calls pass, with no missing records, failures, timeouts or output mismatch. Integer endpoints match the completed certificate studies exactly.

The partial certificate saves approximately 18.5% in the geometric comparison with full certification, while certifying fewer winners on the full panel. Both cost more than numerical exhaustive evaluation. Summing the 360 per-case median times gives 4.686 s, 5.638 s and 3.127 s for partial certificate, full certificate and numerical exact respectively; these sums are not direct end-to-end collection timings. Actual collection wall time is 71.918 s, including input/output, setup and orchestration. The main and certificate timing runs use

Cost ratio	median	geometric mean	equal-family geometric mean
250-rank certificate / full certificate	.826	.815	.820
250-rank certificate / numerical exact	2.063	2.683	2.667
Full certificate / numerical exact	2.713	3.290	3.254

Table 7: Separate three-way timing, same 360 score vectors and five repetitions at $L = 30$. These ratios describe time, so values below one favor the numerator method. They are descriptive finite-panel summaries; no confidence interval was specified for this addendum.

an Apple M1 Pro with 16 GiB RAM, Python 3.12.4, NumPy 1.26.4 and Numba 0.59.1. A single active desktop host limits portability; no cross-platform certificate performance claim is made.

The practical recommendation for these stored scores is therefore simple: use the complete score classes for ordinary numerical evaluation, and request full-grid integer enclosures when a rigorous ideal-target order is required. Partial-grid certification provides a sufficient test and an explicit inconclusive outcome, but this experiment does not establish a better selection algorithm or a faster universal route to that guarantee.

7 Related work and scope of the contribution

VUS and its optimized implementations already exploit static and dynamic anomaly regions (Paparrizos et al., 2022; Boniol et al., 2025). Klein's released implementation sorts support scores and uses cumulative weighted sums (Klein, 2026). Our comparison runs the unchanged pinned source with its actual default grid and separately with exhaustive thresholds; equivalence to a sampled reference is not claimed to be an erroneous exhaustive-integral promise. TimeEval documents deliberate changes in ramps, overlap handling, component counting and integration, while other implementations expose further conventions (TimeEval contributors, 2026; Sylligardos, 2026). Results for those different targets cannot be called approximation error relative to one universal VUS definition. We study the original dynamic right-step PR target only.

The geometry of PR curves and the problems with linear interpolation have been analyzed extensively (Davis & Goadrich, 2006; Keilwagen et al., 2014). Our elementary endpoint envelopes build on that general numerical concern; we make no priority claim for the general concept of an error interval. The specific analysis concerns clipped dynamic recall, component existence, complete ties, and precision growth after saturation. A fixed uniform grid can also severely underestimate early recall increments, a separate effect already documented in the precursor. Both directions of error matter; overestimation is not a universal sign claim.

PAULA bounds precision from sampled ranks under weak or strong precision-monotonicity assumptions while acquiring unknown binary labels (Sabharwal & Sedghi, 2017). Here all labels and scores are available; the enclosure applies to a clipped dynamic integral without assuming monotone precision. The information-access task and target differ, so no annotation-efficiency or performance comparison is implied.

Recent adversarial studies examine whether TSAD metrics reward unskilled detectors or best-of-many reporting (Lyu, 2026). Those questions change reporting behavior or detector construction. Our main empirical question keeps every published score vector and label fixed and changes only threshold integration of one metric contract. MSAD supplies a published model-selection score bank (Sylligardos et al., 2023); re-evaluating its stored scores under our pinned dynamic target does not reproduce or correct its original metric tables, which use a different implementation.

8 Limitations and practical interpretation

Large constructed regret proves possibility. It does not measure the frequency of such cases, invalidate the VUS metric, or imply that its exhaustive winner is optimal for an application's costs. The large finite overestimation theorem uses a tolerance width comparable to series length; the three smaller constructions separately establish substantial regret at maximum widths 30, 100 and 300.

The empirical panel is prospective with respect to this project's metric outcomes, but is not a sample of all anomaly-detection deployments. Exact identity exclusions do not remove every transformed channel or cropped version of a common source recording. Published score quantization is preserved; conclusions concern those stored vectors, not recovered full-precision detector outputs or retrained models. Family-level summaries reduce one source of pseudoreplication without establishing independence of every underlying recording.

Certificate intervals can remain nearly unit width when the retained transcript hides the score ordering that matters. The impossibility example explains why this cannot always be repaired by clever endpoint arithmetic alone. An inconclusive certificate requires additional threshold information or exhaustive evaluation. Soundness is distinct from speed, and integer certification may cost more than a floating-point exhaustive answer. This is a useful explicit tradeoff, not a hidden acceleration claim.

Research assistance and reproducibility. AI tools assisted literature search, mathematical derivation, code, experiment orchestration and drafting. Independent implementations, exact rational checks, unchanged upstream controls, frozen input/source hashes, and preserved failure records test those outputs. Agent reviews are internal checks, not external peer review. The public working draft does not attest to author completion of a submission's review declarations.

References

Paul Boniol, Ashwin K. Krishna, Marine Bruel, Qinghua Liu, Mingyi Huang, Themis Palpanas, Ruey S. Tsay, Aaron Elmore, Michael J. Franklin, and John Paparrizos. VUS: Effective and efficient accuracy measures for time-series anomaly detection. *The VLDB Journal*, 34, 2025. doi: 10.1007/s00778-025-00907-x. URL `https://link.springer.com/article/10.1007/s00778-025-00907-x`.

Jesse Davis and Mark Goadrich. The relationship between precision-recall and roc curves. In *Proceedings of the 23rd International Conference on Machine Learning*, pp. 233–240, 2006. doi: 10.1145/1143844.1143874. URL `https://ftp.cs.wisc.edu/machine-learning/shavlik-group/davis.icml06.pdf`.

Jens Keilwagen, Ivo Grosse, and Jan Grau. Area under precision-recall curves for weighted and unweighted data. *PLOS ONE*, 9(3):e92209, 2014. doi: 10.1371/journal.pone.0092209.

Keras contributors. AUC: Classification metrics, 2026. URL `https://keras.io/api/metrics/classification_metrics/#auc-class`. Threshold discretization and minoring/majoring documentation; inspected 23 October 2026.

Tom Louis Klein. fast-VUS, 2026. URL `https://github.com/tomlouisklein/fast-VUS`. Software, first commit 12 October 2026; audited revision 1bae859f, 16 October 2026.

Youngmin Ko. Sparse vus, version 0.3.0, 2026. URL `https://github.com/mandu5/sparse-vus/releases/tag/v0.3.0`. Public software and working-manuscript precursor, released 21 October 2026; source revision 68d6e6de627a734880e92ecc94d3ffc9eb38a89d.

Zongye Lyu. Did we actually fix it? an independent adversarial stress-test of post-point-adjustment evaluation metrics for time-series anomaly detection, 2026. URL `https://arxiv.org/abs/2607.11969v2`. Version 2, 19 July 2026.

John Paparrizos, Paul Boniol, Themis Palpanas, Ruey S. Tsay, Aaron Elmore, and Michael J. Franklin. Volume under the surface: A new accuracy evaluation measure for time-series anomaly detection. *Proceedings of the VLDB Endowment*, 15(11):2774–2787, 2022. doi: 10.14778/3551793.3551830.

Ashish Sabharwal and Hanie Sedghi. How good are my predictions? efficiently approximating precision-recall curves for massive datasets. In *Proceedings of the 33rd Conference on Uncertainty in Artificial Intelligence*, pp. 411–420, 2017. URL `https://proceedings.mlr.press/r15/sabharwal17a.html`.

Emmanouil Sylligardos. FF-VUS implementation in avusi, 2026. URL `https://github.com/doanthihoaithu/aVUSi`. Software, audited revision facf4362, 23 June 2026.

Emmanouil Sylligardos, Paul Boniol, John Paparrizos, Panos Trahanias, and Themis Palpanas. Choose wisely: An extensive evaluation of model selection for anomaly detection in time series. *Proceedings of the VLDB Endowment*, 16(11):3418–3432, 2023. doi: 10.14778/3611479.3611536. URL `https://github.com/boniolp/MSAD`. Published score archive; repository revision 8141b94eb3d2b1811f81ffcf7bc8aa5dcbd6c204.

TimeEval contributors. Timeeval metrics: Rangeprauc and rangeprvus, 2026. URL `https://timeeval.readthedocs.io/en/latest/api/timeeval.metrics.html`. Documented VUS semantic variations and threshold approximation; inspected 23 October 2026.

VUS contributors. VUS official optimized evaluation source, 2026. URL `https://github.com/TheDatumOrg/VUS/tree/467bace0c103f5c2bdb4900306d82939bcac8ff9/vus`. Pinned revision 467bace0; public get_metrics and RangeAUC_volume_opt default to 250 thresholds.